Genome Pods for Clinical Practice



vito

GHENT UNIVERSITY

IDLab
INTERNET & DATA LAB

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Genome Pods

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Outline

Why?



Genomic Data in Health Care?



Genomic Data in Solid Pods?

What?



How Solid is helpful



Example clinical workflow introduction

How?



“Demo” workflow of Genome Pods UI

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Genome Data

Genome data is big...

~130m



Printed

Human Genome[1]

Ghent Belfry



~91m

Other considerations

- Storage
- Privacy
- Governance
- Formatting

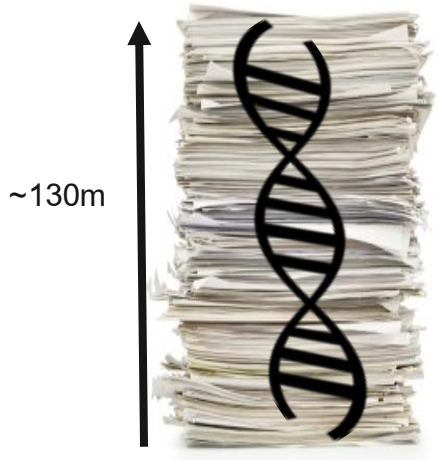
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Genome Data: One Challenge “Solved”

Genome data is big...

Variant Call Format (VCF) file

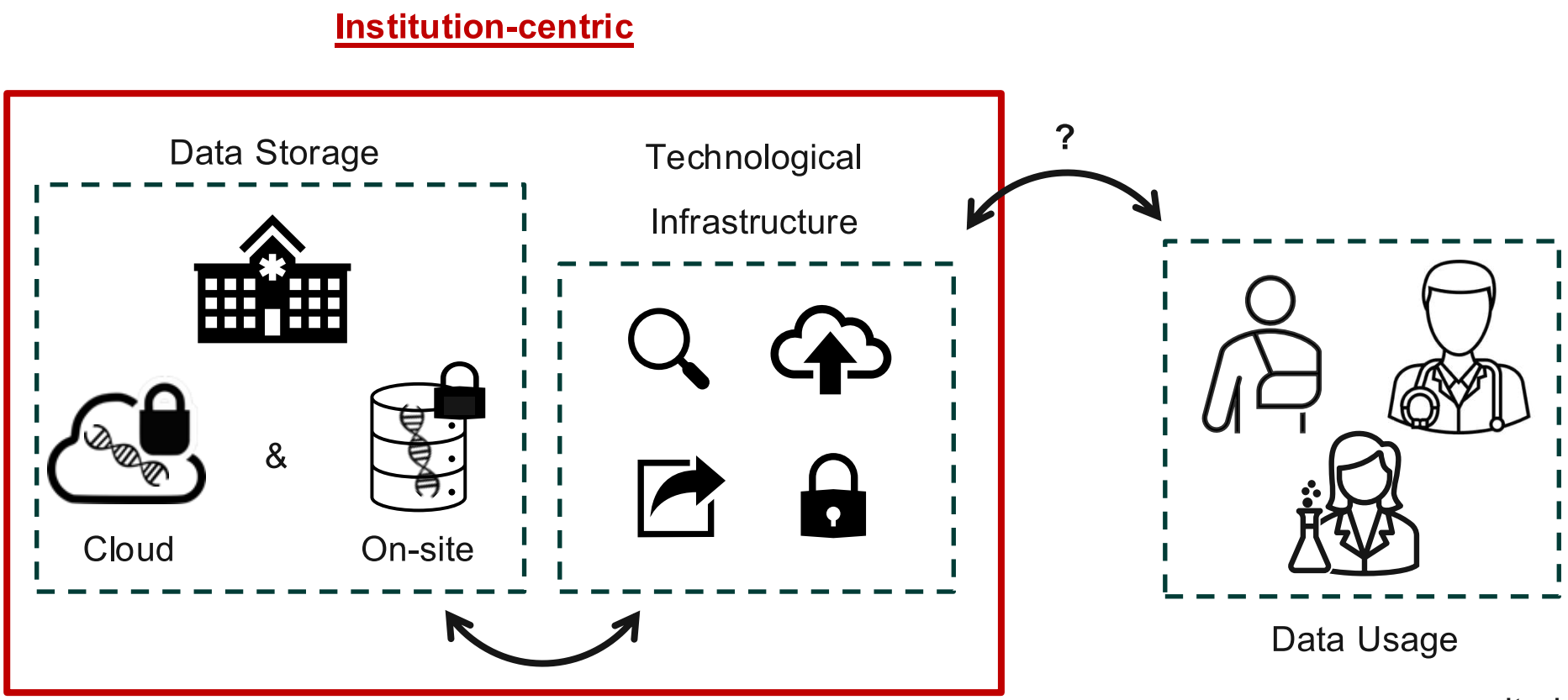
- Individual humans exhibit ~3 million mutations (**~125 MBs**)
- VCF file **records only those differences**



Other considerations: ??

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Genome Data in Health Care



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Scaling Genome Data Usage

Research genomic data sharing initiatives:





State-of-the-art for **clinical** genomic data sharing?

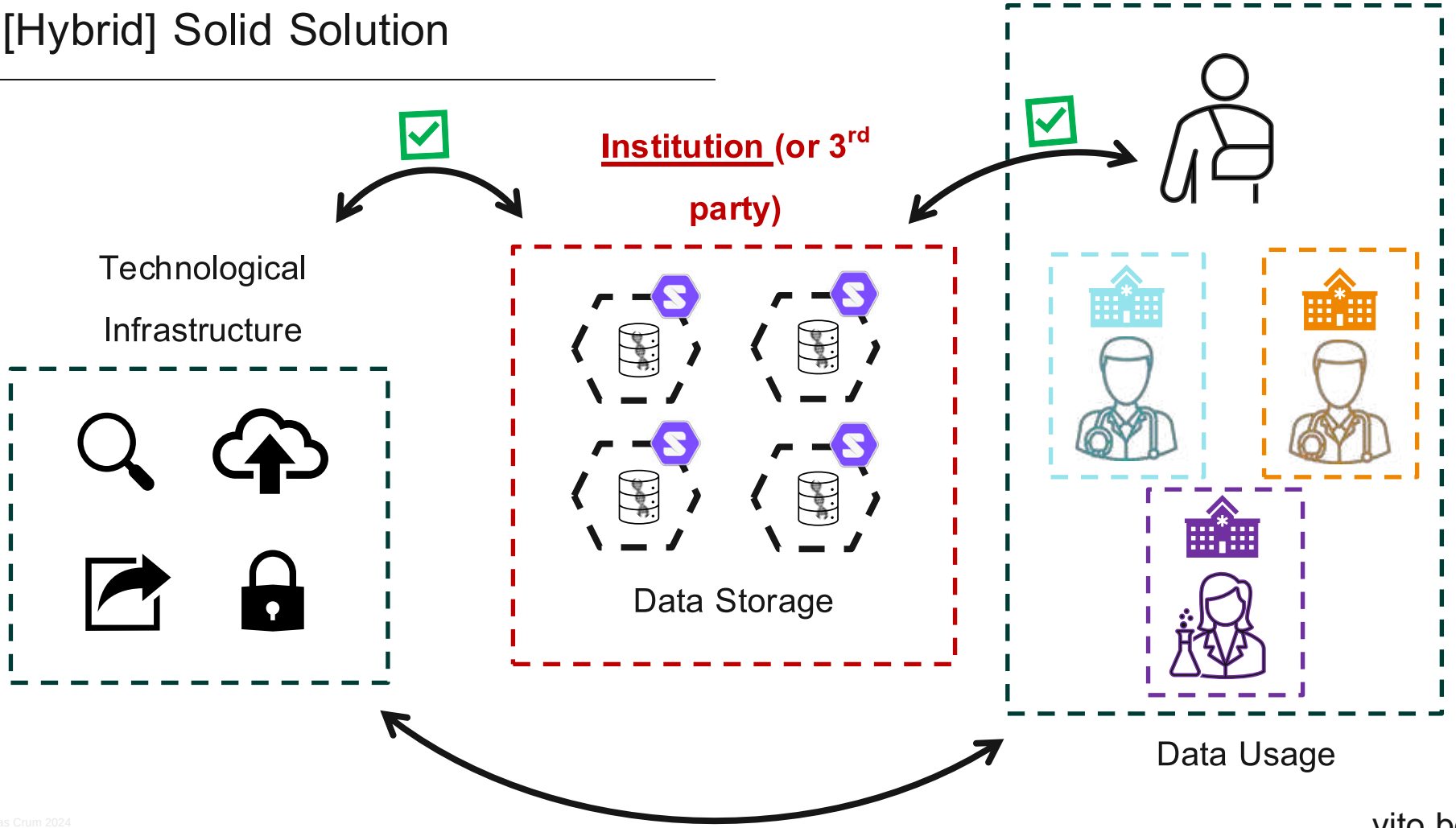
GDPR (EU) / HIPPA (US) *EU Health Data Spaces*



Not much...

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A [Hybrid] Solid Solution



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Solid Technology

- Does not solve ALL problems, but offers an environment where improvements are possible

  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

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Disclaimer & Project Goals

The Genome Pods project is a **RESEARCH** project

It aims to develop a **proof-of-concept**

Framework and **Web Application** for:

- Genomic data storage in Solid Pods
- Granular privacy controls of genomic data
- Genomic data representation in RDF
- Data-linking potential
- Data querying potential (within and across pods)

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Clinical Example

A patient is hospitalized with DVT

(DVT usually merits the perscription of Warfarin[2])

The physician has just received a pharmacogenomic screening

(produced with the patient's VCF file)

The physician uploads the test results to the patient's Genome Pod

The physician makes the test results visible to the patient's primary care physician

(out of network)

The physician queries his other patients for similar tests

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Example Workflow: Physician



Provider Login



App Landing Page



View Patient Data



Add test result to Patient pod



Give read permission to another provider



Query for other patient test results

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Figma example for workflow

Open Here

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Future Directions

Basic Functionalities:

- Genomic data storage in Solid pods
- Data privacy management
- Data Sharing via the web

Experimental research:

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?



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Thank you for listening!

Questions?



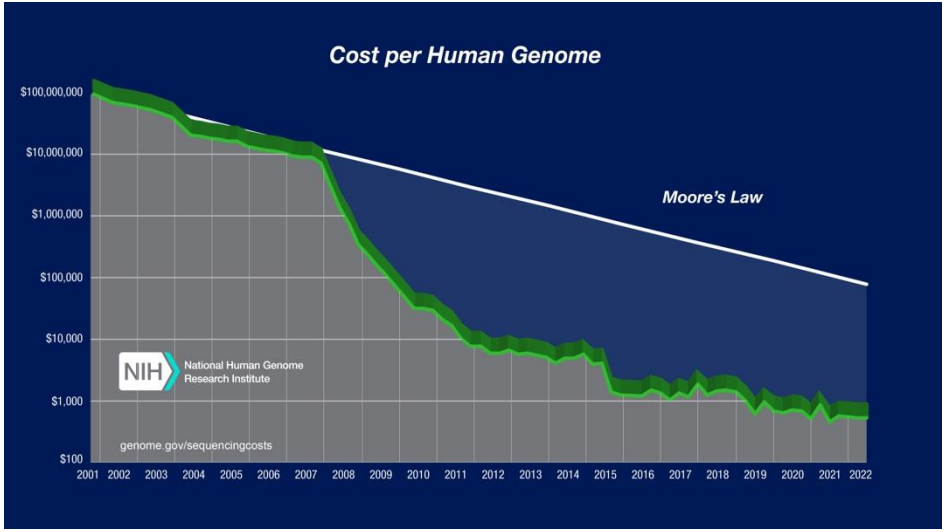
Part of the *PErsoNal Genome QUery IN clinical practice (PENGQUIN)* Ph.D. Project

Funding provided by VITO NV (UG_PhD_2303_contract)

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Genome Sequencing in Health Care

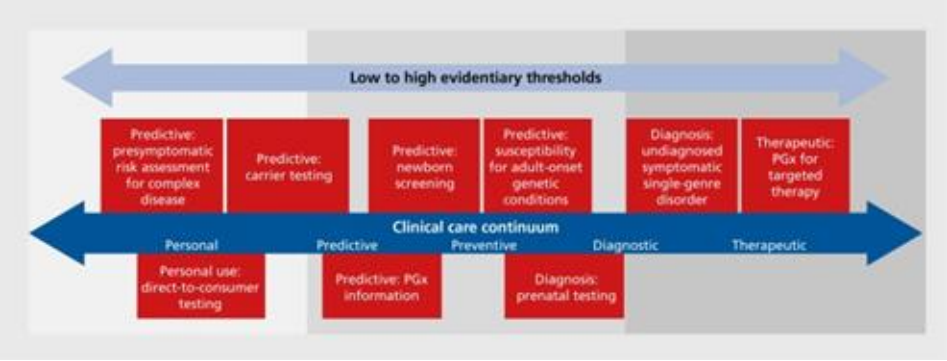
Sequence Generation Costs



Wetterstrand KA. Accessed 12/04/24.

Currently getting close to $100 per genome[4]

Use-cases



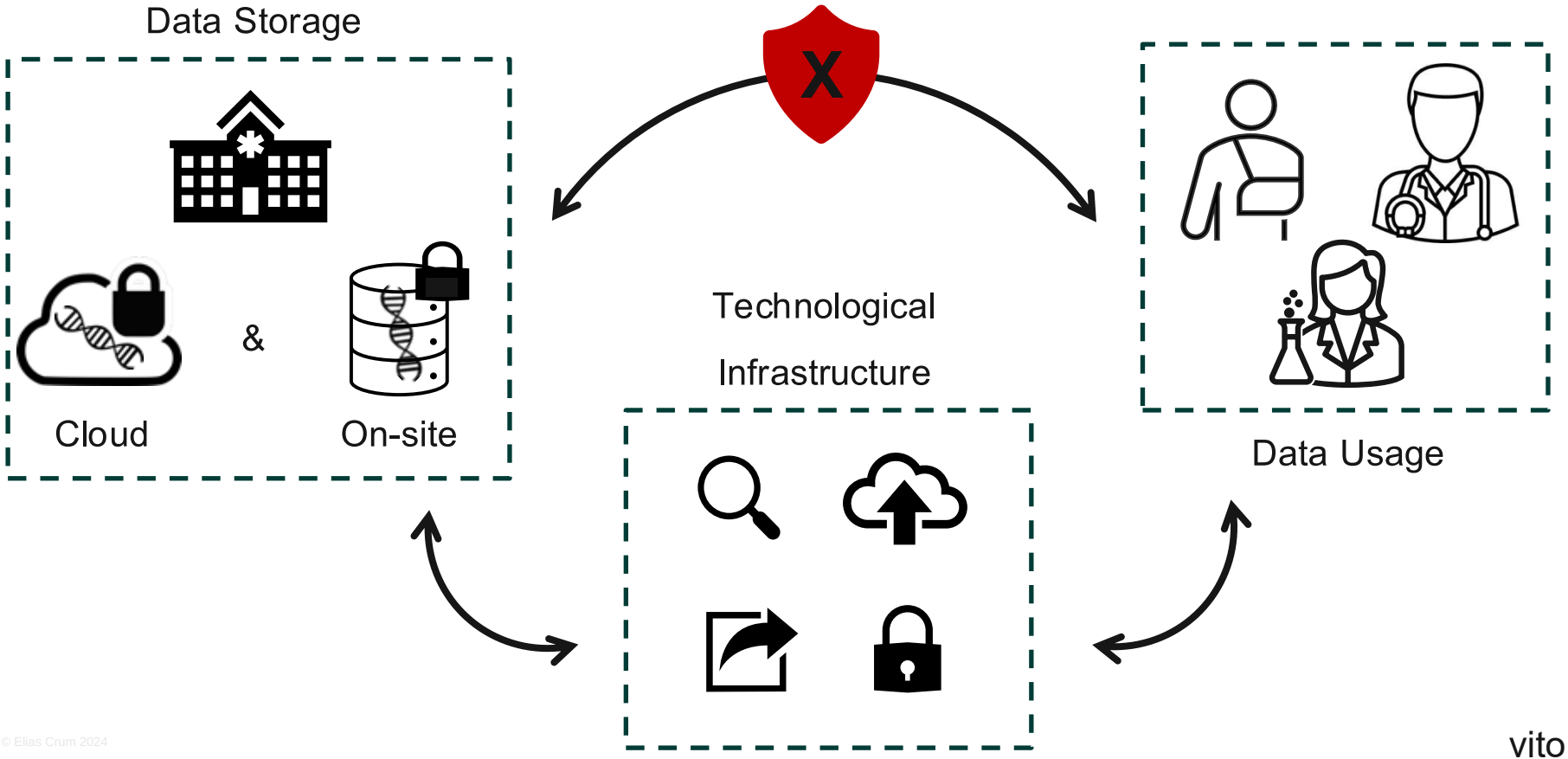
10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

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Genome Data in Health Care



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Implementation Specifics

Solid pod instances	→	Community Solid Server
Genomic Data	→	Publicly available *VCF* data[4]
RDF Conversion	→	Aim to use SPHN RDF vocabulary
Data-linking	→	Actively exploring ontologies / methodologies
Data Querying	→	Custom instance of Comunica

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Sources

1. https://bio4.us/biotrends/human_genome_height.html
2. Tse, G. et al. Genotype-guided warfarin dosing vs. conventional dosing strategies: a systematic review and meta-analysis of randomized controlled trials. 2018. https://doi.org/10.1111%2Fbcp.13621
3. Wetterstrand KA. DNA Sequencing Costs: Data from the NHGRI Genome Sequencing Program (GSP) www.genome.gov/sequencingcostsdata. Accessed 12/04/24.
4. https://www.ultimagenomics.com/publications-and-resources?filter=Literature

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